Fiscal Year:	FY 2021	**Task Last Updated:**	FY 05/18/2022
PI Name:	Dinges, David F. Ph.D.		
Project Title:	Evaluation of SmartSleep Technology for Improving the Efficiency and Restorative Quality of Sleep in Healthy Adults in Order to Mitigate Cognitive Performance Deficits Due to Sleep Restriction and Emergency Awakenings		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	TRISH--TRISH		
Joint Agency Name:		**TechPort:**	Yes
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2018 TRA BRASH1801: Translational Research Institute for Space Health (TRISH) Biomedical Research Advances for Space Health
Start Date:	01/01/2019	**End Date:**	06/30/2021
No. of Post Docs:	1	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	30	**Monitoring Center:**	TRISH
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	NOTE: End date changed to 6/30/2021 per TRISH (Ed., 1/8/21)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Basner, Mathias M.D., Ph.D. (University of Pennsylvania) Garcia Molina, Gary Ph.D. (Philips Electronics North America Corporation) Riedner, Brady Ph.D. (University of Wisconsin) Stahn, Alexander Ph.D. (University of Pennsylvania) Tononi, Giulio M.D., Ph.D. (University of Wisconsin)		
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Task Description:	Sleep in spaceflight is often reduced in duration and of reduced quality, which Earth-based studies show can reduce waking cognitive performance. Sleep medications have been used in spaceflight to promote better sleep quality and longer sleep duration sleep, but these have also been shown to markedly compromise the cognitive performance capability of astronauts when an emergency awakening requires them to function effectively. Thus, there is a need for a technology that can improve sleep quality in space and biologically maximize the performance benefits of limited sleep duration, without unduly affecting the ability of astronauts to awaken abruptly due to an inflight emergency. Studies have found that unconscious sound enhancement of electroencephalogram (EEG) slow waves during sleep can increase subsequent cognitive performance. This approach to improving sleep quality and the cognitive benefits of sleep is now available as "SmartSleep," a new technology that enhances slow waves using inaudible acoustic stimulation during sleep. SmartSleep consists of a headband worn during sleep and a sleep app. This project proposes to evaluate this technology in a controlled, double-blind, laboratory study of non-invasive brain stimulation in healthy normal astronaut-like adults, to evaluate whether slow wave sleep enhancement via the SmartSleep stimulus algorithms could benefit cognitive performance during sleep restriction, and/or decrease the severity and duration of sleep inertia after abrupt awakening from sleep.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The project has unique and innovative impacts for both spaceflight and Earth (e.g., the public) through the evaluation of the SmartSleep headband technology, which is a new, non-invasive, brain stimulation technology to promote sleep depth and quality. The benefit of the research for both, space and Earth, is the potential to reduce the neurocognitive decrements associated with sleep restriction through acoustic stimulation of EEG slow waves during sleep. For space, the SmartSleep headband technology has the potential to benefit astronauts by addressing the need to improve sleep efficiency and quality on spaceflight missions. For Earth, given that adults often sleep less than the suggested 7 hours per night, the SmartSleep headband technology may be a viable at-home device capable of mitigating the neurocognitive decrements associated with short sleep.
Task Progress:	Astronauts must maintain a high level of cognitive performance capability and situational awareness in spaceflight, which depends on their acquiring adequate daily sleep quantity and sleep quality while in space, and on their ability to respond quickly and effectively to emergency events that can occur when they are asleep. Studies have shown that short sleep durations (i.e., less than 6 h per night) are common in astronauts on spaceflight missions aboard the International Space Station (ISS) and Earth-based laboratory studies demonstrate that chronic exposure to the average nightly sleep duration of 6 h on ISS missions cumulatively degrades neurobehavioral functions, with pronounced deficits in vigilant attention and disturbances in mood. Studies have established that astronauts commonly experience short sleep durations, as well as lower quality sleep in spaceflight. Sleep medications have been used in spaceflight to improve sleep quality and sleep duration, yet such medications can also compromise cognitive performance capability following emergent awakenings from sleep. Thus, there is a need for a technology that can biologically maximize the performance benefits of limited sleep duration, without adversely affecting the ability of astronauts to awaken abruptly due to an inflight emergency. Studies have found that acoustic enhancement of electroencephalogram (EEG) slow waves during sleep can increase subsequent cognitive performance. This approach to improving sleep quality and the cognitive benefits of sleep is now available as "SmartSleep," a new technology that enhances slow waves using inaudible acoustic stimulation during sleep. SmartSleep consists of a headband worn during sleep and a sleep app. The project utilized this technology in a controlled, double-blind, laboratory study of non-invasive brain stimulation in healthy normal astronaut-like adults, to evaluate whether slow wave sleep enhancement via the SmartSleep stimulus algorithms can benefit cognitive performance during sleep restriction, and/or decrease the severity and duration of sleep inertia after abrupt awakening from sleep. The project had three objectives: Objective 1: Evaluate whether slow wave sleep enhancement via the SmartSleep stimulus algorithms benefits cognitive performance during a period of chronic sleep restriction as commonly observed during spaceflight (Deliverable 1). We will assess cognitive performance with the Cognition test battery (See Basner et al. PMCID: PMC4691281) across a range of cognitive domains and across a number of high fidelity spatial cognition tasks and spaceflight-relevant operational tasks including Robotic On-Board Trainer (ROBoT); Six Degrees of Freedom Docking Simulator (6DF); and Lunar Landing Simulation. Objective 2: Investigate whether SmartSleep slow-wave enhancement decreases performance decrements induced by sleep inertia after an emergent awakening (Deliverable 2). Objective 3: Identify the best modality of SmartSleep acoustic stimulation in terms of slow-wave enhancement and cognitive performance among the following three different stimulation techniques: Continuous Fixed Interval (1 Hz, inter-tone interval stimulation), Block (5 seconds on versus 5 seconds off), and In-Phase Adjustable (constant stimulation with tones to be delivered during each upstate of the slow wave) [Deliverable 3]. The project proposed the following specific aims: Specific Aim 1: Determine whether EEG slow-wave sleep can be enhanced via closed-loop auditory stimulation of EEG slow waves in healthy adult subjects using SmartSleep Headband technology, when sleep is restricted in duration to 5h per night over four consecutive nights. Specific Aim 2: Determine whether post-sleep cognitive performance functions sensitive to sleep restriction can be enhanced when SmartSleep closed-loop auditory stimulation is used on sleep-restricted nights relative to a sham SmartSleep control condition. Specific Aim 3: Identify the acoustic stimulation modality with maximal benefits for EEG slow-wave enhancement and daytime cognitive performance, when sleep is restricted in duration to 5h per night over 4 consecutive nights. Specific Aim 4: Investigate systematic effects of consecutive nights of sleep restriction on EEG slow-wave enhancement, daytime cognitive performance, emotion recognition and emotional valence, and stress reactivity. Specific Aim 5: Assess potential adverse effects of SmartSleep utilization during sleep restriction, especially related to sleep inertia after emergent awakening. Data collection has concluded and yielded a final sample of n=8 of a planned total N=12 participants that completed the

	7-day laboratory protocol (i.e., n=48 subject nights completed in the laboratory, out of a projected total of 72 nights). The reduction in sample size was a result of the restrictions on human subjects' research due to the COVID-19 pandemic. On March 16, 2020, the University of Pennsylvania (UPenn) Perelman School of Medicine halted all non-life sustaining research. On July 13, 2020, the project was approved by UPenn to resume research at 50% capacity in accordance with university mandated procedures to ensure the health and safety of participants and research personnel. Following research resumption approval, the project resumed data collection and n=2 participants completed the study protocol, yielding the final sample of n=8 participants. The SmartSleep headband device recorded polysomnography (PSG) data across all laboratory nights, which produced a total of 48 nocturnal PSG recordings. Participants wore the SmartSleep headband device without reported discomfort or adverse events and participants did not report hearing audio tones during wakefulness or sleep. Furthermore, participants completed all neurocognitive and spaceflight-relevant operational tasks.
Bibliography Type:	Description: (Last Updated: 06/30/2026)
Abstracts for Journals and Proceedings	Dinges DF, Basner M, Kaizi-Lutu M, Mange A, Stahn AC, Tononi G, Riedner B, Garcia-Molina G, Jones CW "Evaluation of SmartSleep technology for improving the efficiency and restorative quality of sleep in healthy adults in order to mitigate cognitive performance deficits due to sleep restriction and emergency awakenings." 2020 NASA Human Research Program Investigators' Workshop, Galveston, Texas, January 27-30, 2020. Abstracts. 2020 NASA Human Research Program Investigators' Workshop, Galveston, TX, January 27-30, 2020. , Jan-2020
Abstracts for Journals and Proceedings	Jones CW, Basner M, Kaizi-Lutu M, Stahn AC, Tononi G, Riedner B, Garcia-Molina G, Dinges DF. "Evaluation of SmartSleep technology for improving the efficiency and restorative quality of sleep in healthy adults in order to mitigate cognitive performance deficits due to sleep restriction and emergency awakenings." 2021 NASA Human Research Program Investigators' Workshop, Virtual, February 1-4, 2021. Abstracts. 2021 NASA Human Research Program Investigators' Workshop, Virtual, February 1-4, 2021. , Feb-2021
Articles in Peer-reviewed Journals	Tu D, Basner M, Smith MG, Williams ES, Ryder VE, Romoser AA, Ecker A, Aeschbach D, Stahn AC, Jones CW, Howard K, Kaizi-Lutu M, Dinges DF, Shou H. "Dynamic ensemble prediction of cognitive performance in spaceflight." Sci Rep. 2022 Jun 30;12:11032. https://doi.org/10.1038/s41598-022-14456-8 ; PubMed PMID: 35773291; PubMed Central PMCID: PMC9246897 , Jun-2022